MGU-UGP BSc (HONS) MICROBIOLOGY

III SEMESTER

DSC A - MG3DSCMBG200- GENERAL MICROBIOLOGY

(2024 ONWARDS)

Time: 1.5 Hours **Max. marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any 10 questions.

Each question carries 2 marks.

1. What is numerical taxonomy? [K][1]
2. Explain phylogenetic classification. [U][1]
3. What are fimbriae and pili? [K][1]
4. Write down the structure of Gram-positive flagella. [U][1]
5. Explain the chromophore and auxochromegroups.[U][2]
6. Write down the advantages of negative staining. [U][2]
7. List the objectives of fixation. [K][2]
8. Discuss about Mcintosh jar. [U][2]
9. Define sterilisation. [K][3]
10. What is cold sterilisation? [K][3]
11. Classify antibiotics based on mode of action. [U][3]
12. Explain thermal death time. [U][3]

(10x2=20)

Part B

Short Essay Questions

Answer any 4 questions.

Each question carries 5 marks.

13. SummariseBergey's Manual of Systematic Bacteriology. [U][1]
14. List the features of a good disinfectant. [K][3]
15. Compare the features of Gram-positive and negative cell walls. [U][1]
16. Describe anaerobic culture methods. [U][2]
17. Summarise the process of sporulation. [U][1]
18. Explain the principle of the autoclave. [A][3]

(4x5=20)

Part C

Long Essay Question

Answer any 1question

Each question carries 10 marks

19. Draw and explain the structure of gram Gram-negative cell wall with a neat diagram. [U][1]
20. Explain applications and uses of moist heat sterilisation. [A][3]

(10x1=10)